Basic concepts of radiation detection

Detectors can be roughly divided in

Thermal detectors

Absorbed power (quasi blackbody)
↓
Temperature difference ΔT
↓
output signal

- The response is proportional to the absorbed power
- does not depend on λ

Pros:

Direct measurement of input power
Wide spectral sensitivity
IR sensitivity up to FAR IR (THz)

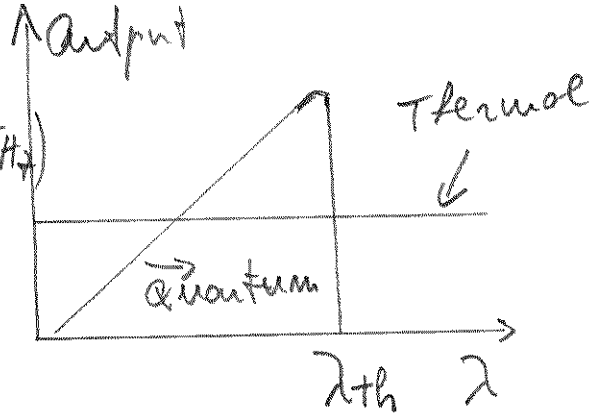
Cons:

Low sensitivity

Quantum detectors

Photons interactions in one to one events with electrons
↓
Generation of free carriers
↓
output signal (current)

The response is:
- proportional to the rate absorbed photons
- The response depends on λ
- A threshold exists



Pros:
- High sensitivity up to single photon

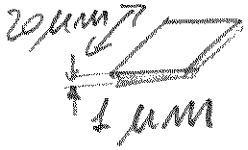
Cons:
- $out = out(\lambda)$ (spectrum a priori)
- Wavelength threshold
- Require cooling beyond MIR

POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO

Rough estimation of the sensitivity of a thermal detector.

- Microbolometer

$$P \rightarrow \Delta T \rightarrow \Delta R \rightarrow \Delta V$$



silicon

$M = 10^{-12}$ Kg Mass

$C = 8{,}6 \cdot 10^{-10}$ J/K Thermal capacitance

noise equivalent thermal difference

How many photons are required to produce $\Delta T = NETD = 50\,mK$

$\lambda = 10\,\mu m$ (Peak of a Blackbody $\simeq 37°C$)

$E_{ph} = 2 \cdot 10^{-20}$ J

$$N_{ph} = \frac{C \cdot NETD}{E_{ph}} \simeq 10^9 \text{ photons}$$

POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITEC

Figures of merit of radiation detectors

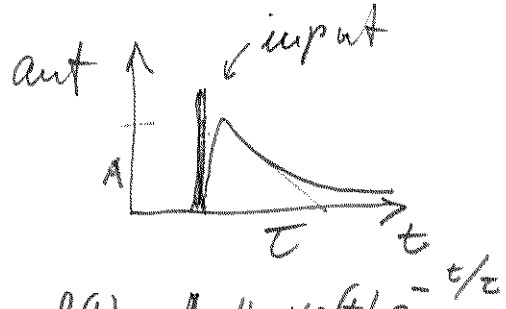
- **Responsivity**

Ratio of the output signal versus the input power.

$$R(\lambda) = \frac{S_{out}[A]}{P_{in}[W]} \quad ; \quad R(\lambda) = \frac{S_{out}[V]}{P_{in}[W]}$$

Responsivity does not correspond to sensitivity

Useful to design the front-end of the detector



$$h(t) = A \cdot \mathrm{Heavis}(t)\, e^{-t/\tau}$$

- **Frequency response**

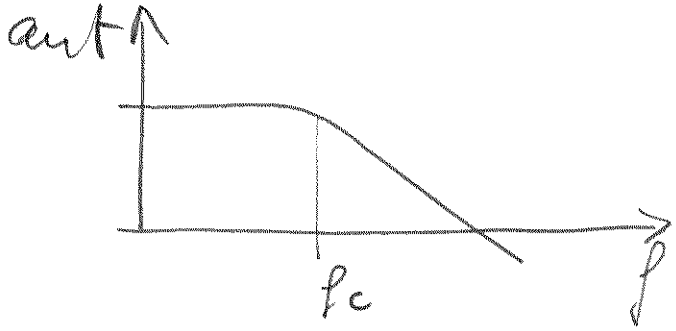
Typically is limited by a time constant $\tau = R_L C$

$$H(s) = \frac{A\tau}{1+s\tau} \qquad H(s) = \mathcal{L}[h(t)]$$

$$R_\lambda(f) = |H(f)| = \frac{A\tau}{\sqrt{1+4\pi^2 f^2 \tau^2}} = \frac{R_\lambda(0)}{\sqrt{1+4\pi^2 f^2 \tau^2}}$$

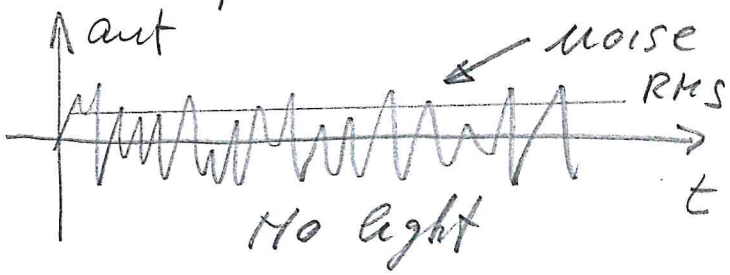
Cutoff frequency

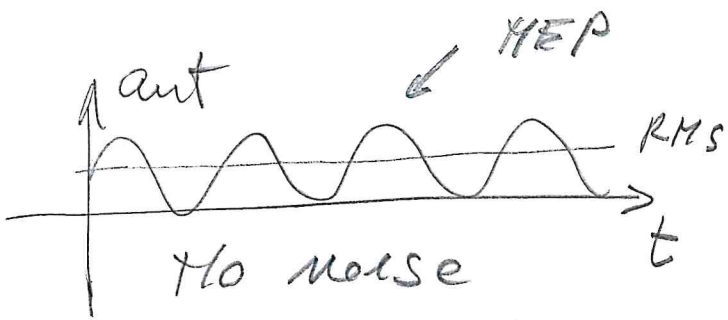
$$f_c = \frac{1}{2\pi\tau} \qquad R_\lambda(f_c) = \frac{R_\lambda(0)}{\sqrt{2}}$$

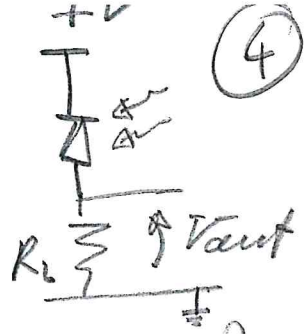


POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITEC

Noise equivalent power







NEP = optical power in sinusoidal regime that would produce in a noise free detector a RMS signal = the RMS noise measured from the detector with NO signal

$$NEP \longrightarrow S/N = 1$$

$$\left[N_{RMS} = \sqrt{\frac{1}{T}\int_0^T N^2(t)\,dt} \right]$$

Bandwidth

Noise power $\propto A \cdot \Delta f$

$$N_{RMS} \propto \sqrt{A}\,\sqrt{\Delta f}$$

equal for NEP

$$NEP_\lambda = P_{[S/N=1]} = \frac{S_{RMS}}{R_\lambda} = \frac{N_{RMS}}{R_\lambda} \propto \sqrt{A}\,\sqrt{\Delta f}.$$

$$\left[R_\lambda = \frac{S_{out}}{P_{in}} \qquad P_{in} = \frac{S_{out}}{R_\lambda} \right]$$

$$NEP^*_\lambda = \frac{NEP}{\sqrt{A \cdot \Delta f}}$$

$$D_\lambda = \frac{1}{NEP_\lambda} \qquad D^*_\lambda = \frac{1}{NEP^*_\lambda}$$

POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITE